

WEB102 Prework

Challenge 1 of 7: Getting Started

Background

About Git

Throughout this course, you will be using Git to keep track of your work, prepare assignments for submission, and collaborate with others. Git is a version control system that allows programmers to save versions of their work over time, much like the version history in Google Suite products. Git also allows multiple people to collaborate on the same code without overwriting each others' work. The system will keep track of changes and allow you to merge them together in an organized way.

Using Git to Keep Track of Changes

At its core, Git keeps track of changes to files and reconciles differences between different versions of the same file. For example, you just forked a repository created by CodePath. A repository is simply a location where code is stored. Forking created your own copy of the code separate from the one that CodePath owns. Git is complex and it will take time to fully understand. If some of the commands below seem obscure, just know that it feels that way to everyone at first and over time they will start to make more sense.

It's common to interact with Git using the terminal or command prompt. If this is your first time using the terminal, it can seem overwhelming! The terminal is a way for developers to communicate with the operating system. Throughout this course, you'll become more comfortable as you continue to practice with this environment. For Git specifically, commands should start with git, followed by one of the operations Git can perform.

Challenge Steps

Step 1: Set Up Your Environment

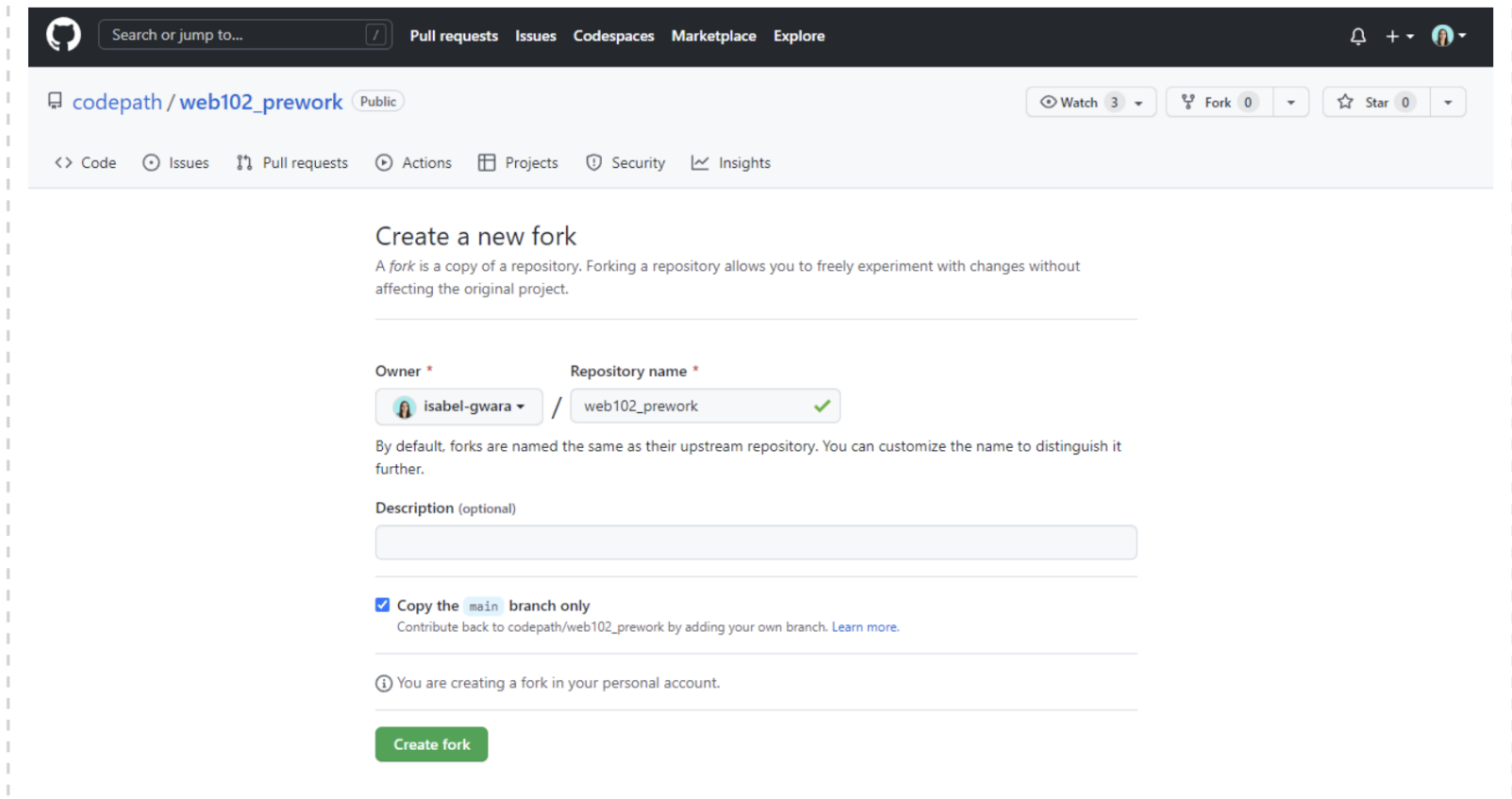
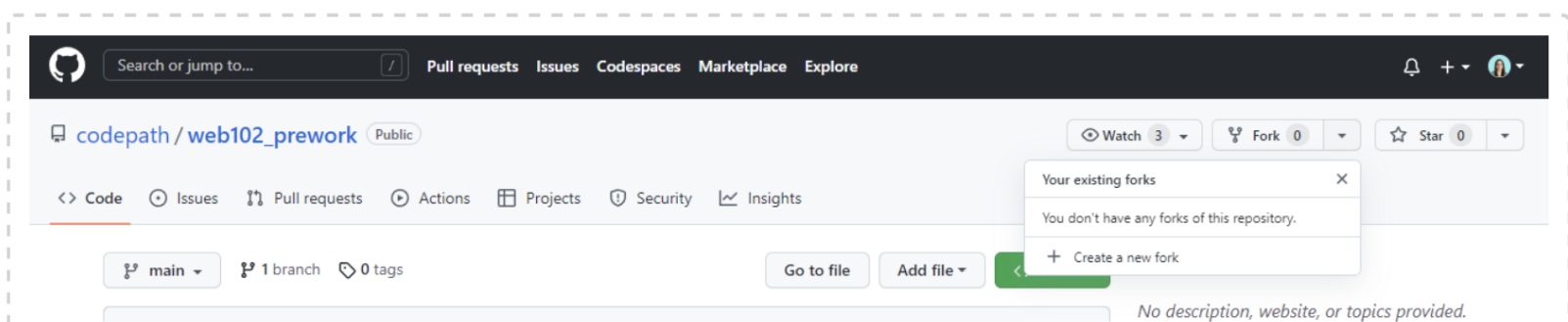
- **Download and install VS Code** on your computer
 - VS Code is the Integrated Development Environment (IDE) we will be using throughout the course.
- **Download and install Git.**
 - Git is the version control system we will use throughout the course.
- (Optional) **Install the suggested VS Code extensions**
 - GitHub Pull Requests and Issues: allows you to easily review and manage pull requests and issues as you collaborate
 - Live Server: allows you to start a local development server with a live reload feature. Useful for plain HTML/CSS/JS projects, though this functionality is built into Vite, which we will use throughout the course.

Here are some resources you may find helpful for setting up VS Code and GitHub:

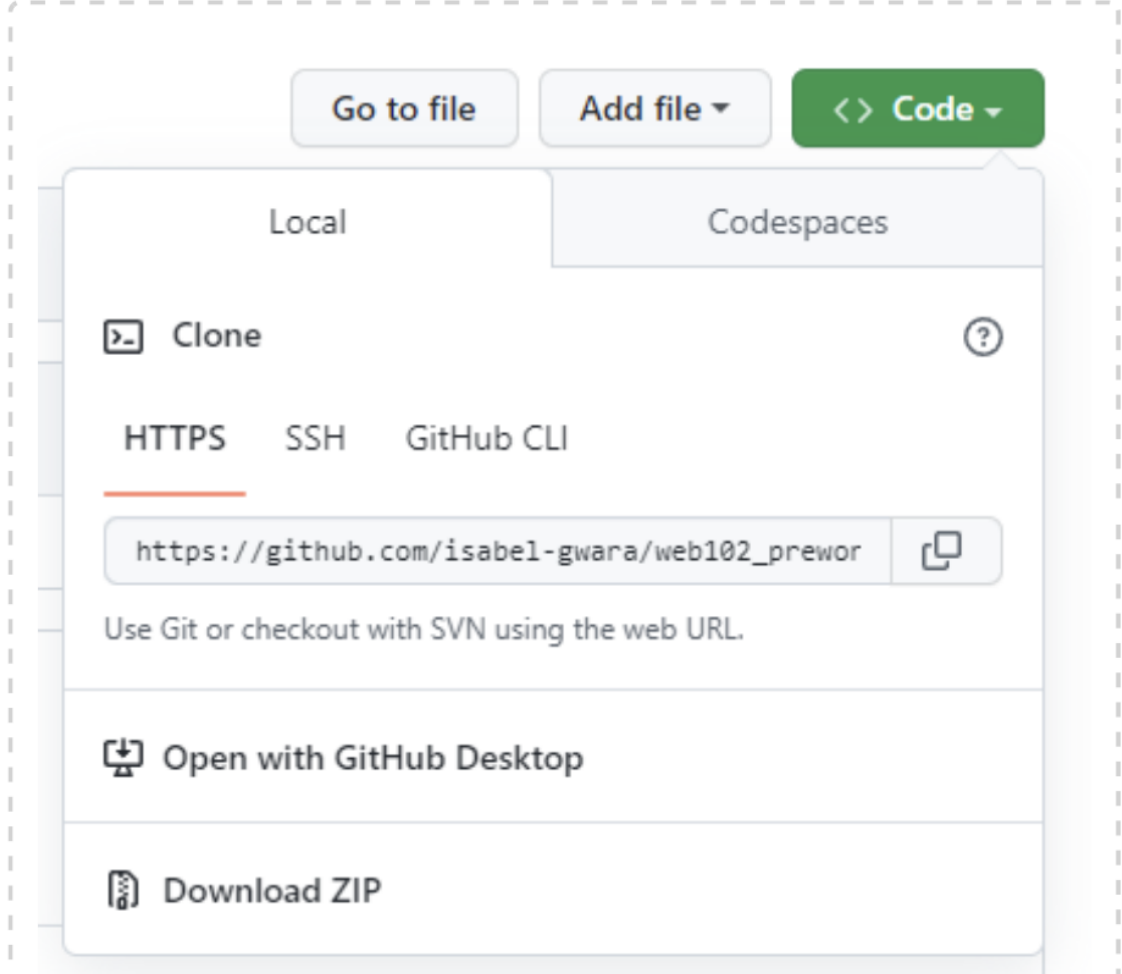
- [GitHub Walkthrough](#)
- [Visual Studio Code Walkthrough](#)
- [Live Server VS Code Extension](#)
- [Windows Dev Environment Set-up Instructions](#)

Step 2: Set Up Your Project

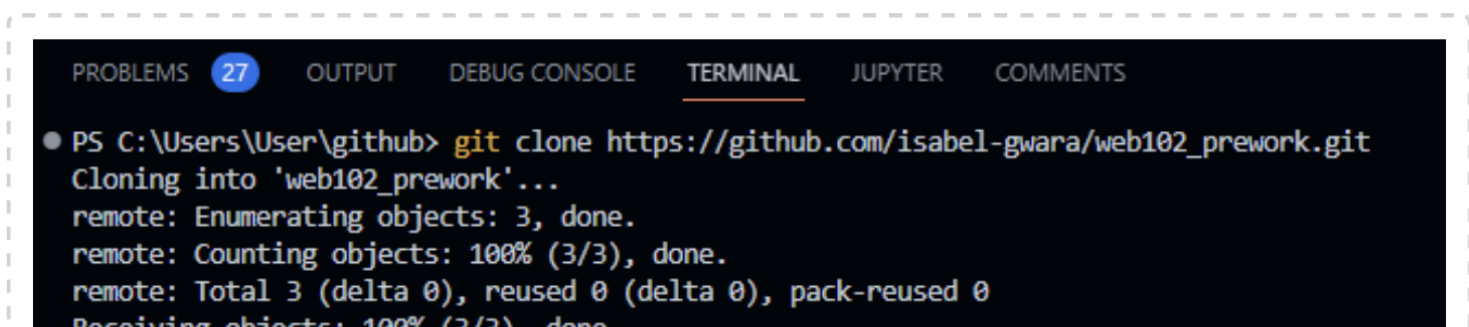
- **Sign into a GitHub account.** You should use the same account for this prework assignment that you used to apply to CodePath. Sign into your GitHub account [here](#).
- **Fork the [WEB102 Prework Repository](#) on Github.**



- **Clone the fork you created.**
 - Navigate to the fork you just created, select the Green "<> Code" button in the top right and copy the provided Clone link



- Once in VS Code, open the command prompt or terminal using the keyboard shortcut **cmd + `** or **ctrl + `**
- In the terminal, use the **git clone** command along with the URL to your repository (you copied this in the previous step!). It should look something like this:
 - **git clone https://github.com/YOUR_USERNAME/web102_prework.git**
- Once you receive the done message, use the **ls** command (list files), to confirm you created a local repository. You should now see a list of all files in your current directory, including a new folder containing a local copy of all of the files in your GitHub repository.



- In VS Code, select **File -> Open Folder** and navigate to open your cloned repository folder on your computer.

Step 3: Learn about Git

Take about 8 minutes to read through "[An Easy Beginner's Guide to Git](#)" before you proceed. Remember that you may want to open this resource in a new tab or window; if this window closes, you'll need to re-enter your  **Secret Key**.

Make sure to read the following sections:

- Committing Files to the Local Repository Ignoring Files in Commits
- Git Branches
- Merging branches
- Create a Remote Repository to Share with Others
- Moving Code Between Local and Remote Repository
- Cloning a Remote Repository

Step 4: Git Quiz!

To move on to the next section, complete the short Git quiz below. Each keyword for the correct answer will form part of the  **Secret Key**.

1. Which command starts a fresh new repository?

1. **git clone <link-to-repository>** (keyword: SKEW)
2. **git init** (keyword: OOZE)
3. **git push origin master** (keyword: ALLY)
4. **git --version** (keyword: FLAP)

2. What happens when executing the command git add **index.js**?

1. A copy of the file **index.js** is created in the working directory (keyword: crib)
2. A copy of the file **index.js** is added to the remote repo (keyword: harp)
3. The file **index.js** is moved from the local repo to the staging area (keyword: best)
4. The file **index.js** is moved from the working directory to the staging area (keyword: dive)

3. What is the process for adding files to the local repository called?

1. committing (keyword: TRAP)
2. pushing (keyword: LIME)
3. forking (keyword: FUSE)
4. cloning (keyword: GAZE)

4. What happens when executing the command **git branch bug-fixes** then **git checkout bug-fixes**?

1. Code from the branch **bug-fixes** will be pushed to the remote repository. (keyword: zest)
2. All commits on the branch **bug-fixes** will be listed in the terminal. (keyword: walk)
3. A new branch **bug-fixes** will be created and then moved to. (keyword: pine)
4. The branch **bug-fixes** will be updated with all recent commits. (keyword: play)

Moving on to Challenge 2

Create the  Secret Key by combining all four of your Git Quiz answers together. For example, if all the correct answers in this quiz were A, the  **Secret Key** would be **SKEWcribTRAPzest**. Double check your spelling and capitalization!

Use your  **Secret Key** to unlock the next PDF file. If your  **Secret Key** is correct, you will move onto the next step!

Reminders about  **Secret Keys**:

- Keys for the Offline PDF version of prework **are case sensitive!**
 - (For example, "SEA_MONSTER" is NOT the same as "sea_monster")

Secret keys can be any combination of letters, numbers, and symbols.